

Supplementary Material for: PGCASurv: A Prior-Guided Cross-Attention Framework for Dynamic Survival Model with Longitudinal Data

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SI. PAIRED EFFECT-SIZE ANALYSIS FOR PERFORMANCE EVALUATION

To complement the p -value-based comparisons in the main manuscript, fold-wise paired effect sizes were computed using the same five cross-validation folds. For C-index, IAUC, and td-AUC, the paired difference was defined as PGCASurv minus the comparator, whereas for IBS, the sign was reversed and defined as the comparator minus PGCASurv. Thus, positive values consistently indicate better performance of PGCASurv across all metrics. The confidence intervals were computed as two-sided 95% confidence intervals for the paired differences. The reported p -values are based on one-sided paired t -tests, consistent with the main manuscript. Stars denote statistical significance: $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$. In each table, the strongest external baseline denotes the external baseline with the best mean performance for the corresponding dataset, landmark, and metric.

A. Global Evaluation

Table S1 summarizes the paired effect sizes against the strongest external baseline in the global evaluation. The results show that PGCASurv achieves clear positive differences for some comparisons, such as C-index and IAUC on PBC2. At the same time, several confidence intervals include zero, and a few mean paired differences are negative. These results indicate that, although PGCASurv is statistically competitive overall, the magnitude of improvement over the strongest baseline is dataset- and metric-dependent, and some differences should be interpreted as modest or uncertain rather than uniformly large.

B. Dynamic Evaluation

Table S2 reports the corresponding paired effect sizes for landmark-based dynamic evaluation on the PBC2 dataset. The estimates are positive for most landmark-metric combinations, suggesting that PGCASurv generally improves over Landmark-Cox in this dynamic setting. However, the two-sided 95% confidence intervals are relatively wide and often include zero, especially at later landmarks. This pattern indicates that the horizon-specific improvements are subject to uncertainty,

which is consistent with the modest absolute differences reported in the main manuscript. The negative mean difference for IBS at 428 days further suggests that the advantage of PGCASurv is not uniform across all prediction horizons and metrics.

SII. MISSINGNESS SENSITIVITY ANALYSIS

To evaluate robustness to feature-level missingness, we conducted a missingness sensitivity analysis by injecting element-wise missing values under a missing completely at random (MCAR) mechanism.

A. Setup

We performed this analysis on five real-world longitudinal survival datasets: PBC2, AIDS, FRAMINGHAM, EPILEPSY, and LIVER. Specifically, for each dataset, we randomly masked entries in the dynamic covariate matrix with missing rate $r \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$. For each missing rate, we applied the same MCAR masking procedure independently to both the training and test splits. We then followed the same cross-validation, model training, and evaluation pipeline as in the main experiments. Performance was assessed using C-index, integrated Brier score (IBS), and Integrated time-dependent AUC (IAUC).

B. Results

For each missing rate, we report the mean and standard deviation computed across the 5 cross-validation folds; in Figures S1–S3, shaded bands indicate ± 1 standard deviation around the fold-mean curve.

For Fig. S1, we observe an overall pattern in which most datasets show either a mild decrease or near-stable C-index as the missing rate increases. For the LIVER, EPILEPSY, and AIDS datasets, only very slight decreases are observed, and the shaded bands remain relatively narrow, indicating that PGCASurv maintains stable discrimination performance across the considered missingness levels on these datasets. In FRAMINGHAM, a clearer monotonic decreasing trend is present, while the shaded bands are not substantially widened, suggesting a consistent reduction in discrimination as more information is missing. For PBC2, the C-index exhibits a noticeable drop at a missing rate of 0.3 with a substantially wider shaded band; the performance then partially recovers at

TABLE S1: Paired effect-size analysis against the strongest external baseline for each dataset and metric in the global evaluation.

Metric	Dataset	Strongest external baseline	$\bar{\Delta}$	95% CI	One-sided p -value
C-index	AIDS	CRDR-Surv	0.0106	[-0.0036, 0.0249]	0.0535
	EPILEPSY	MATCH-Net	0.0106	[-0.0068, 0.0281]	0.0832
	FRAMINGHAM	MATCH-Net	0.0035	[-0.0072, 0.0143]	0.2055
	LIVER	CRDR-Surv	0.0017	[-0.0044, 0.0078]	0.2422
	PBC2	CRDR-Surv	0.0695	[0.0373, 0.1018]	0.0020**
IBS	AIDS	Cox Landmarking	-0.0343	[-0.0586, -0.0101]	0.9914
	EPILEPSY	Cox Landmarking	-0.0002	[-0.0181, 0.0178]	0.5093
	FRAMINGHAM	MATCH-Net	0.0012	[-0.0006, 0.0030]	0.0711
	LIVER	MATCH-Net	0.0109	[-0.0106, 0.0324]	0.1152
	PBC2	MATCH-Net	0.0328	[-0.0113, 0.0768]	0.0540
IAUC	AIDS	Cox Landmarking	0.0061	[-0.0714, 0.0835]	0.4190
	EPILEPSY	MATCH-Net	0.0063	[0.0003, 0.0122]	0.0216*
	FRAMINGHAM	MATCH-Net	-0.0075	[-0.0153, 0.0002]	0.9728
	LIVER	CRDR-Surv	-0.0030	[-0.0107, 0.0047]	0.8308
	PBC2	MATCH-Net	0.0507	[0.0279, 0.0734]	0.0017**

TABLE S2: Paired effect-size analysis against the strongest external baseline for each landmark and metric in the dynamic evaluation on the PBC2 dataset.

Landmark	Metric	Strongest external baseline	$\bar{\Delta}$	95% CI	One-sided p -value
161 days	C-index	LandmarkCox	0.0793	[-0.0151, 0.1736]	0.0400*
	IBS	LandmarkCox	0.0155	[-0.0043, 0.0353]	0.0094**
	td-AUC	LandmarkCox	0.0678	[-0.0184, 0.1541]	0.0472*
305 days	C-index	LandmarkCox	0.0266	[-0.0403, 0.0935]	0.1659
	IBS	LandmarkCox	0.0118	[-0.0268, 0.0505]	0.2214
	td-AUC	LandmarkCox	0.0348	[-0.0453, 0.1150]	0.1470
428 days	C-index	LandmarkCox	0.0627	[-0.1346, 0.2601]	0.2136
	IBS	LandmarkCox	-0.0193	[-0.1064, 0.0678]	0.7140
	td-AUC	LandmarkCox	0.0385	[-0.1971, 0.2741]	0.3367

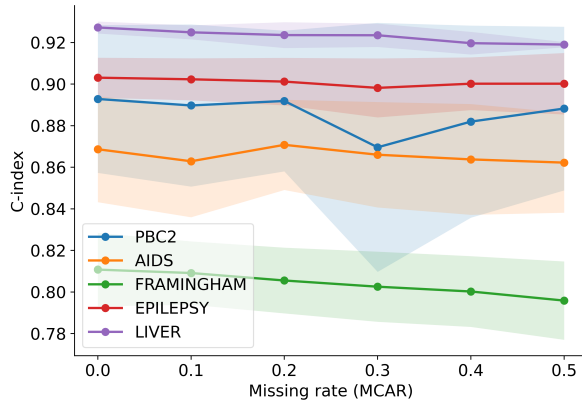


Fig. S1: C-index versus missing rate under element-wise MCAR missingness for five datasets.

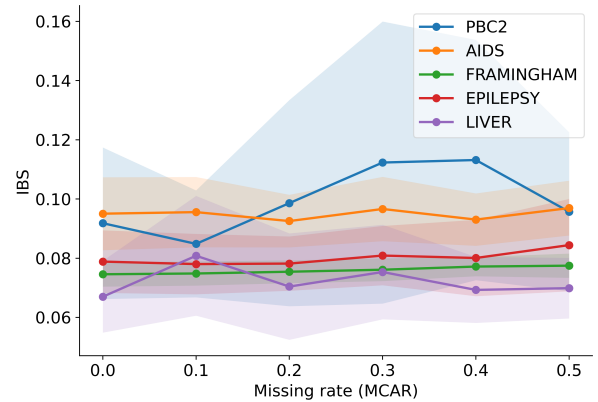


Fig. S2: IBS versus missing rate under element-wise MCAR missingness for five datasets.

missing rates of 0.4–0.5. This pattern suggests increased fold-to-fold variability at intermediate missingness levels, which may contribute to the observed non-monotonic behavior.

For Fig. S2, IBS values across the five datasets fluctuate within a relatively narrow range overall, while the stability differs noticeably by dataset. On FRAMINGHAM and EPILEPSY, PGCASurv shows relatively stable IBS across missingness levels, with only mild changes as the missing rate increases. On AIDS and LIVER, the curves exhibit some

fluctuations but remain generally stable, and the shaded bands are relatively narrow. In contrast, PBC2 shows larger IBS variation, with a clear increase in the mean IBS around missing rates of 0.3–0.4 accompanied by a substantially wider shaded band.

Fig. S3 shows a pattern that is largely consistent with Fig. S1: for most datasets, IAUC remains stable or decreases only mildly as the missing rate increases from 0 to 0.5. For the four datasets other than PBC2, PGCASurv yields

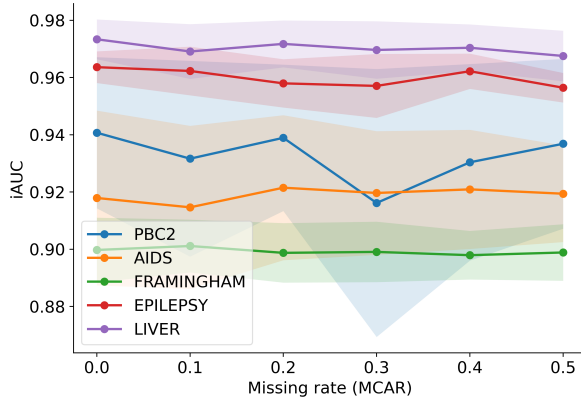


Fig. S3: IAUC versus missing rate under element-wise MCAR missingness for five datasets.

relatively stable IAUC curves across missingness levels, with only modest fluctuations. In contrast, on PBC2, IAUC exhibits a noticeable drop at a missing rate of 0.3 along with a substantially wider shaded band, followed by a recovery at higher missing rates.

III. RISK RECLASSIFICATION ANALYSIS

A. AIDS Dataset

Following the same risk reclassification protocol described in the main text, we compared the static (w/o dynamic) model with the full PGCASurv model on the AIDS dataset and identified subjects whose binary risk labels (high vs. low) changed when incorporating longitudinal information.

For each baseline risk stratum defined by the static model, we compared subjects whose risk status changed under PGCASurv with those whose status remained unchanged, where risk labels were defined at the pre-specified horizon $\tau = 10$ years. Table S3 summarizes the results on the AIDS dataset. Among the evaluated covariates, CD4 shows evidence of a group difference ($p = 1.37 \times 10^{-2}$). In contrast, gender does not show clear evidence of a difference at conventional significance levels, and drug and AZT exhibit no evidence of group differences in this comparison. Overall, this analysis provides a coarse-grained characterization of covariates associated with risk reclassification on the AIDS dataset under our reclassification protocol. Notably, CD4 also yields the smallest p -value among the tested covariates, suggesting that it may play a comparatively more prominent role in distinguishing subjects who undergo risk reclassification at the 10-year horizon. This aligns with clinical and epidemiologic evidence that CD4 cell count is a key marker of HIV disease severity and is strongly associated with progression to AIDS and mortality [1].

B. Epilepsy Dataset

For the Epilepsy dataset, we performed the risk reclassification analysis at the pre-specified horizon $\tau = 1095$ days (3 years) by comparing the static (w/o dynamic) model with the full PGCASurv model.

Table S4 summarizes the results on the Epilepsy dataset. Under the tests, gender yields the smallest p -value ($p =$

TABLE S3: Covariate t -tests for patients whose risk stratification changes between the w/o-dynamic model and PGCASurv on the AIDS dataset.

Covariate	p -value
CD4	1.37×10^{-2}
gender	8.31×10^{-2}
drug	7.14×10^{-1}
AZT	7.55×10^{-1}
prevOI	8.85×10^{-1}

3.25×10^{-2}), suggesting a potentially more prominent association with risk reclassification than the other covariates in this coarse-grained screening. This is broadly consistent with clinical evidence that sex is associated with differences in epilepsy presentation and may also relate to differential response to antiseizure medications [2], [3].

Overall, these findings provide an approximate characterization of covariates associated with risk reclassification on the Epilepsy dataset under our 3-year reclassification protocol.

TABLE S4: Covariate t -tests for patients whose risk stratification changes between the w/o-dynamic model and PGCASurv on the Epilepsy dataset.

Covariate	p -value
gender	3.25×10^{-2}
drug	9.53×10^{-2}
dose	1.07×10^{-1}
age	2.76×10^{-1}
disab	8.81×10^{-1}

C. Liver Dataset

For the Liver dataset, we performed the risk reclassification analysis at the pre-specified horizon $\tau = 5$ years by comparing the static (w/o dynamic) model with the full PGCASurv model in table S5. Proth shows evidence of group differences under the tests ($p = 1.94 \times 10^{-4}$). Notably, proth yields the smallest p -value among the tested covariates, suggesting a comparatively more prominent association with risk reclassification in this coarse-grained screening. Overall, these findings provide an approximate characterization of covariates associated with risk reclassification on the Liver dataset under our 5-year reclassification protocol.

TABLE S5: Covariate t -tests for patients whose risk stratification changes between the w/o-dynamic model and PGCASurv on the Liver dataset.

Covariate	p -value
proth	1.94×10^{-4}
Trt	9.50×10^{-1}

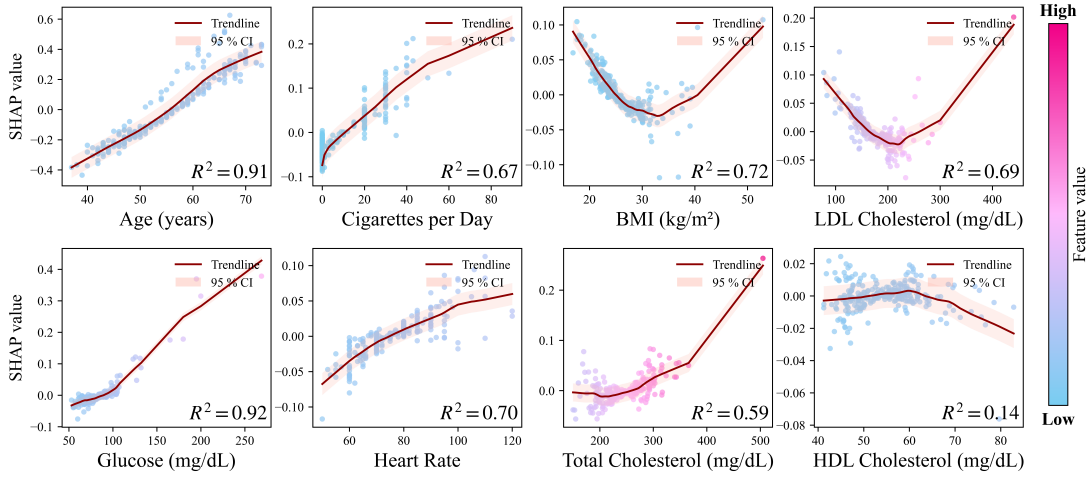


Fig. S4: SHAP dependence plots at the landmark time $t = 1957$ days on the Framingham dataset using PGCASurv. The dependence patterns are broadly consistent with those at $t = 2014$ days shown in the main text.

D. PBC2 Dataset

For the PBC2 dataset, we performed the risk reclassification analysis at the pre-specified horizon $\tau = 730$ days (2 years) by comparing the static (w/o dynamic) model with the full PGCASurv model.

Table S6 reports the results on the PBC2 dataset. Among the evaluated covariates, serBilir and SGOT yield the smallest p -values, followed by albumin, edema, and alkaline. Several additional variables also show evidence of group differences, including hepatomegaly, spiders, histologic, and ascites. In contrast, other covariate do not show clear evidence of differences between groups in this comparison. This pattern is consistent with established PBC prognostic models, where serBilir and albumin are core variables for long-term risk stratification [4], [5].

Overall, this analysis provides a coarse-grained characterization of covariates associated with risk reclassification on the PBC2 dataset under our 2-year reclassification protocol.

TABLE S6: Covariate t -tests for patients whose risk stratification changes between the w/o-dynamic model and PGCASurv on the PBC2 dataset.

Covariate	p -value	Covariate	p -value
serBilir	1.25×10^{-5}	ascites	5.71×10^{-3}
SGOT	1.32×10^{-5}	prothrombin	2.48×10^{-2}
albumin	5.47×10^{-5}	serChol	2.84×10^{-2}
edema	3.83×10^{-4}	platelets	1.81×10^{-1}
alkaline	6.88×10^{-4}	age	2.83×10^{-1}
hepatomegaly	2.42×10^{-3}	sex	7.57×10^{-1}
spiders	2.66×10^{-3}	drug	7.94×10^{-1}
histologic	5.49×10^{-3}		

SIV. ADDITIONAL SHAP ANALYSES

Supplementary Fig. S4 shows the SHAP dependence plots at the landmark time $t = 1957$ days on the Framingham

dataset. The overall shapes are highly consistent with those at $t = 2014$ days reported in the main text (Fig. ??(a)), suggesting that the marginal effects of key covariates on the predicted risk remain stable during this early to intermediate period. In particular, AGE exhibits an approximately linear increasing trend, indicating higher predicted risk with older age. BMI and LDLC show an approximate U-shaped pattern, suggesting elevated risk at both low and high values of these markers. GLUCOSE displays a monotone increasing relationship, with higher glucose levels corresponding to larger SHAP values. In contrast, the HDLC-related pattern becomes more pronounced at the later landmark time ($t = 4082$ days; Fig. ??(b)), consistent with the time-varying emphasis highlighted in the main text.

SV. ADDITIONAL OPTIMIZATION DETAILS

This section summarizes the hyperparameter search space used for the Optuna-based Bayesian optimization of the proposed model.

TABLE S7: Hyperparameter search space used for Optuna-based Bayesian optimization.

Hyperparameter	Type	Search space
d_{model}	categorical	{64, 128}
$d_{\text{stat,hidden}}$	categorical	{64, 128}
$d_{\text{dyn,hidden}}$	categorical	{64, 128}
n_{heads}	categorical	{2, 4}
Dropout rate	continuous	[0.0, 0.5]
λ_{TTE}	log-uniform	$[10^{-3}, 5 \times 10^{-1}]$
λ_{smooth}	log-uniform	$[10^{-5}, 10^{-2}]$
Learning rate	log-uniform	$[10^{-4}, 5 \times 10^{-3}]$
Weight decay	log-uniform	$[10^{-6}, 10^{-3}]$

TABLE S8: Mean wall-clock training time (in seconds) per cross-validation experiment on five datasets.

Models	AIDS	Epilepsy	Framingham	Liver	PBC2
Dynamic-DeepHit	2.98	5.48	34.68	8.68	4.92
MATCH-Net	13.59	11.02	147.89	14.65	9.38
STRAFE	5.78	3.59	34.63	4.54	1.58
CRDR-Surv	6.91	21.50	24.76	16.90	11.13
Cox Landmarking	0.21	0.19	0.36	0.15	0.14
PGCASurv(proposed)	5.56	11.76	68.06	9.92	8.35

SVI. HARDWARE CONFIGURATION AND RUNTIME ANALYSIS

To quantify computational cost, we report the average wall-clock training time required to complete one full cross-validation run for each model–dataset combination (Table S8), together with the hardware and software configuration of the workstation used in our experiments (Table S9).

TABLE S9: Hardware and software configuration of the workstation used in our experiments.

Category	Specification
Operating System	Windows 11
CPU	AMD Ryzen 5 6600H
Memory	16 GB RAM
GPU (Discrete)	NVIDIA GeForce RTX 3050
Python	3.11.9
PyTorch	2.6.0+cu124
CUDA	12.4

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